Makhaola Nkhahle – Interface lead

CS3520 · LIMO Simulator · Milestone M2 · Team Binary Brains
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I build everything the user sees and clicks in the browser simulator.

My files

File	What it does
index.html	The page layout: editor, machine-code table, register table, and empty panels for the pipeline.
src/style.css	Colours, the phone and laptop layouts, and the colours for each instruction field.
src/app.js	Connects the buttons to the assembler, fills the tables, and shows errors.
docs/M2-design.md §7	Wireframes for the eleven panels.

My work

- Program editor with an example dropdown, an “Open .limo” button, and an Assemble button.
- Machine-code table: each 32-bit word is split into its fields (opcode, rd, rs1, rs2, funct, imm/disp, unused). Each field has its own colour, a gap between fields, a name on hover, and a colour legend.
- Register table for r0–r15 with the Sesotho names (lefela, mokokotlo, aterese ea khutlo).
- Simple error messages with the line number.
- One column on phones, two columns on laptops.
- Colour is never the only clue: spacing, hover text, and labels go with every colour.
- Wireframes for the whole simulator (controls, datapath, control signals, hazards, pipeline chart, statistics, registers, memory, freshman-mode narration) for M3 and M4 to build on.